FREE FRAMEWORK

AI Stack Decision Framework

A systematic approach to choosing the right AI tools — and avoiding $50k mistakes

📊 Evaluation Matrix

💰 Cost Comparison

✅ POC Checklist

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Why Most Teams Get This Wrong

Common Mistakes

- ✕ **Hype-driven decisions** — Choosing tools based on what's trending on Twitter
- ✕ **Vendor lock-in** — Picking proprietary solutions without exit strategy
- ✕ **Over-engineering** — Using complex solutions for simple problems
- ✕ **Ignoring costs** — Not factoring in API costs, infrastructure, and maintenance

The 5-Step Framework

Used by 100+ teams to make confident AI tool decisions

1

Define Your Problem

Before looking at tools, answer these questions:

QUESTION	EXAMPLE ANSWER
What task are you automating?	Customer support ticket classification
What's your latency requirement?	< 500ms (real-time) or batch okay?
What's your accuracy threshold?	95%+ for this use case
What's your monthly budget?	$500/month for this feature
Do you need custom training?	No, pre-trained model should work
What's your data sensitivity?	Contains PII, needs data residency

2

Map Tool Categories

Map your problem to these categories:

LLM APIS
GPT-4, Claude, Gemini, Llama

VECTOR DATABASES
Pinecone, Weaviate, Chroma, PGVector

EMBEDDING MODELS
OpenAI, Cohere, Sentence Transformers

ORCHESTRATION
LangChain, LlamaIndex, custom code

MONITORING
Langfuse, Weights & Biases, custom

DEPLOYMENT
Vercel, AWS, GCP, self-hosted

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Evaluation Matrix

Score each candidate tool on these dimensions (1-5):

DIMENSION	WEIGHT	TOOL A	TOOL B	TOOL C
Performance (accuracy/speed)	30%			
Cost (per 1K requests)	25%			
Ease of integration	20%			
Vendor stability	15%			
Customization options	10%			
Weighted Score	100%			

4

Build vs Buy Decision Tree

```
Do you have ML expertise in-house?
├── No → Start with API-based solutions
└── Yes → Do you need <10ms latency?
    ├── Yes → Consider self-hosted models
    └── No → Is data privacy critical?
        ├── Yes → Self-hosted or private cloud
        └── No → API solutions acceptable
```

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Proof of Concept Checklist

Before committing, validate with a 2-week POC:

- ☐ Integrate with your actual data
- ☐ Measure end-to-end latency
- ☐ Calculate projected costs at scale
- ☐ Test error handling and edge cases

- [ ] Evaluate documentation and support
- [ ] Check community size (GitHub stars, Discord)
- [ ] Review vendor roadmap and stability

4 Common Patterns

Pattern 1: Simple Classification

~$0.002/request

🔧 OpenAI GPT-3.5 + Function Calling ✓ Clear categories, < 100 categories

Pattern 2: RAG (Retrieval Augmented Generation)

~$0.005/query

🔧 LlamaIndex + OpenAI Embeddings + Pinecone ✓ Need to reference your documents

Pattern 3: Real-time Chat

~$0.001/message

🔧 Claude 3 Haiku + WebSocket + Redis ✓ Conversational interfaces

Pattern 4: Batch Processing

~$1-2/hour GPU

🔧 Self-hosted Llama 3 + Celery + PostgreSQL ✓ High volume, cost-sensitive

Cost Comparison (per 1M tokens)

PROVIDER	INPUT	OUTPUT	BEST FOR
GPT-4o	**$5.00**	$15.00	Complex reasoning
Claude 3.5 Sonnet	**$3.00**	$15.00	Coding tasks
Claude 3 Haiku	**$0.25**	$1.25	Simple tasks
GPT-4o Mini	**$0.15**	$0.60	Cost/performance
Llama 3 (self-hosted)	**~$0.05**	~$0.05	Requires GPU management

Red Flags to Avoid

1. **No API documentation** — You'll waste weeks reverse engineering
2. **No pricing page** — Enterprise sales means $$$$
3. **No SOC 2** — Security teams will block it
4. **< 100 GitHub stars** — May be abandoned
5. **No versioned API** — Breaking changes will break you

Your Decision Template

Copy this for your next evaluation:

```
Project: [Name]

Requirements
- Latency: [X ms]
- Accuracy: [X%]
- Budget: [$X/month]
- Scale: [X requests/day]

Candidates Evaluated
1. [Tool A] - Score: X/25 - [Decision]
2. [Tool B] - Score: X/25 - [Decision]
3. [Tool C] - Score: X/25 - [Decision]

Chosen Solution
[Tool] because [reasons]

POC Results
- Latency: X ms ✓/✗
- Accuracy: X% ✓/✗
- Cost: $X/month ✓/✗

Decision
[Proceed / Pivot / Abandon]
```

Need Help With Your AI Stack Decision?

I offer hands-on support for teams building AI systems:

AI Architecture Audit

2-week comprehensive review ($3,000)

RAG System Build

Production-ready implementation ($8,000)

Fractional AI Engineer

Ongoing advisory ($10k/month)

Book a free 15-minute consultation: calendly.com/chinadbo/30min

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This framework is based on 50+ AI implementations. Results may vary based on your specific use case.

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